



Digital-ABB

Infrastructure Hygiene & Physical Layer Protection

Sample Health & Hygiene Audit - Rack, Cable, Airflow & Physical-Layer Risk

Audit Positioning: This sample report is designed to support audit readiness and internal facility reviews. Digital-ABB benchmarks environmental observations against recognized infrastructure hygiene practices and references NFPA 75 fire-safety considerations and ISO 14644-1 Class 8 environmental expectations where applicable. Final compliance determinations remain the responsibility of the client, certifying authority, insurer, or OEM.

Audit Summary

Client / Site	Sample Client - Server Room A
Service Type	Health & Hygiene Audit / Visual Condition Review
Environment	Rack row, cable area, intake/exhaust airflow paths
Method	ESD-aware, non-disruptive observation and documentation
Output	Risk notes, particle observations, photos, and action plan

Financial Risk Formula

Potential cost of 1 hour downtime	\$5,600+ example baseline*
Quarterly preventive maintenance	\$400 example starting point
Business logic	One avoided incident can offset multiple maintenance cycles.

*Downtime cost varies by business. This example is for discussion only and should be replaced with the client's internal downtime estimate.

Laser-Verified Particle Observation

Metric	Before Cleaning	After Cleaning	Observation
PM2.5	___ ug/m3	___ ug/m3	Record with handheld meter
PM10	___ ug/m3	___ ug/m3	Record with handheld meter
Visible dust load	Low / Med / High	Low / Med / High	Photo documented
Airflow concern	Yes / No	Yes / No	Rack intake / exhaust

Before & After Visual Proof

Before Photo: particle meter screen, rack intake, cable area, or heatsink/fan zone.	After Photo: same location after controlled cleaning and dust removal.
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Insert before photo here	Insert after photo here
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Risk Findings Checklist

- Airflow restriction at intake/exhaust zones
- Dust accumulation on rack doors, vents, cable bundles, and equipment-adjacent surfaces
- Fiber-area contamination risk and connector-adjacent dust exposure
- Signs of thermal stress, fan loading, or airflow bottlenecks
- Recommended maintenance window for dry fan/heatsink cleaning when equipment can be powered down

Actionable Recommendations

Priority	Recommendation	Timing
High	Remove rack/cable dust from airflow pathways using ESD-aware methods.	Immediate / pilot
Medium	Schedule powered-down dry fan and heatsink cleaning for selected devices.	Annual or bi-annual
Medium	Document before/after particle readings and photos for internal review.	Each service
Low	Set quarterly rack hygiene cadence for high-traffic or dust-sensitive spaces.	Quarterly